

and was easy to understand. Later, B was rewritten and renamed as the C programming language by Dennis Ritchie. The C language is still used to write system level programs and time critical applications. It is easy to understand, which is why all universities consider C as the primary programming language for students.

How UNIX became the world's most popular OS

In November 1973, Ken Thompson and Dennis Ritchie formally presented the UNIX operating system to the outside world for the first time at a conference called 'The Symposium on Operating Systems Principles' at Purdue University, USA. A number of students, researchers, professors and representatives from private organisations were present at the event.

There was also one special person present that day, who changed the direction of UNIX development — Prof. Bob Fabry from the University of California, Berkeley, was responsible for taking UNIX to Berkeley.

After the conference, UNIX became public and the buzzword for computing power. Newspapers and magazines featured stories about this new computing system. UNIX was made available to the government, the military and to universities on a licensed basis.

Prof. Bob Fabry obtained a copy of UNIX for US\$ 99 in order to cut costs when setting up the campus computing resources for research at the university. At that time, a UNIX licence also came with the entire source code, and researchers could modify and extend it. Graduate student Bill Joy, along with researchers and hackers from Berkeley, started working on making changes to the original UNIX.

The word had spread, and the Berkeley-modified version of UNIX became more and more popular amongst research scientists, educational institutes and universities.

In early 1977, Joy released Berkeley UNIX under the name BSD (Berkeley Software Distribution), while AT&T

continued developing UNIX under the names 'System III' and later 'System V'. Till 1980, AT&T Bell Labs developed multiple versions of UNIX for internal use with different hardware compatibility.

At that time, there were two UNIX development tracks running in parallel — one at AT&T Bell Labs and another by the academic community, led by the Computer Systems Research Group (CSRG) at the University of California, Berkeley.

In the late 1980s through the early 1990s, the wars between these two major strains continued to rage. After many years, each variant adopted many of the key features of the other. Commercially, System V won the standards' battle (getting most of its interfaces into the formal standards), and most hardware vendors switched to AT&T's System V.

In the 1980s, many companies started building their own OSs, with UNIX as a base. They would obtain a UNIX copy either from AT&T Bell Labs or from Berkeley Software Distribution, and adapt it for their own hardware. In this way, many popular UNIX based OSs such as FreeBSD, NetBSD, Sun Solaris, Zenix, IBM-AIX and HP-UX came into existence.

In 1982, Sun Microsystems developed a UNIX operating system called SunOS for its workstations and server computing systems. It was based on the Berkeley Software Distribution UNIX, from version 1 to version 5.0. Later versions were based on AT&T Bell Labs' UNIX System V Release 4 and were marketed under the brand name Solaris, as proprietary operating system software. In June 2005, Sun Microsystems released most of the code base under the CDDL (Common Development and Distribution License) and founded the OpenSolaris open source project. Sun wanted to build a developer and user community around the software. But after the acquisition of Sun Microsystems by Oracle in January 2010, the latter decided to discontinue the OpenSolaris distribution and development model.

In 1984, HP came up with HP-UX, which was based on AT&T Bell Labs' UNIX version System V.

In 1986, technology giant IBM obtained UNIX version System V with 4.3BSD-compatible extensions from AT&T Bell Laboratories. The company made it compatible with IBM architecture-based hardware systems and started selling it under the brand IBM-AIX (Advanced Interactive eXecutive). It became the most popular UNIX commercial distribution in the 90s and even later. Originally, AIX was released for the IBM RT PC RISC workstation. It now supports a wide variety of hardware platforms, including the IBM RS/6000 series as well as POWER and PowerPC-based systems, IBM System i, System/370 mainframes, PS/2 personal computers and the Apple Network Server.

The final UNIX release from Berkeley was in 1995 — the 4.4BSD-Lite Release 2, after which the Computer Systems Research Group was dissolved, and the development of BSD at Berkeley ceased. Since then, several variants based directly or indirectly on 4.4BSD-Lite (such as FreeBSD, NetBSD, OpenBSD and DragonFly BSD) have been maintained by the open source community.

In 1980, a UNIX-like operating system called NeXTSTEP was developed by Steve Jobs' company called NeXT. The NeXTSTEP kernel was based on the Mach kernel which was originally developed by Carnegie Mellon University (the Mach kernel was based on BSD UNIX). Later, NeXT was acquired by Apple Inc. This OS got rebranded as the MacOS, which is now a proprietary operating system of Apple Inc. and there is no direct relation with UNIX.

Linux enters the picture

The obvious question now is that if UNIX contains everything — one branch for enterprises, driven by AT&T, and another university developed BSD UNIX — how did Linux enter the picture and become more popular?